BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 080 · Trace Tables and Algorithm Refinement

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Name one useful item to record when correcting a failed test.

Q2. Which access rights should a user have?
A. Full administrator rights
B. All other users' rights
C. No login checks
D. Only those they need

Q3. “A printer driver stores only the user's website passwords.” Correct this. Explain why.

Q4. A program reads names from a file and saves results. Correct both claims. Explain why. (a) readLine() writes a line. (b) writeLine(text) only shows text on screen.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Trace table — Manually records changes to variables and printed outputs as code executes.

Accumulator — A variable updated repeatedly to store a running total.

Expected output — The result that should occur according to the problem requirements.

Refinement — Use discrepancies in the trace to revise instructions and retest.

Test case — An input and expected outcome used to check correctness.

Bug — A defect in the program.

New learning

Trace table: Manually records changes to variables and printed outputs as code executes. Consider this in a points tally.

Accumulator: A variable updated repeatedly to store a running total. Consider this in a shop receipt sum.

Expected output: The result that should occur according to the problem requirements. Consider this in a school class score.

Refinement: Use discrepancies in the trace to revise instructions and retest. Consider this in a three-item total.

Worked example

A points tally needs trace table and accumulator. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Trace table [1]; Manually records changes to variables and printed outputs as code executes. [1].

3. Accumulator [1]; A variable updated repeatedly to store a running total. [1].

4. Give a specific point and a linked reason for each feature.

We Do · Practise with guidance

Guided 1

In a shop receipt sum, explain the purpose of trace table. [2 marks]

Worked method: Trace table [1]; Manually records changes to variables and printed outputs as code executes. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Guided 2

A school class score needs accumulator. Explain its role. [2 marks]

Worked method: Accumulator [1]; A variable updated repeatedly to store a running total. [1].

Guided 3

A three-item total uses expected output and refinement. Explain both. [4 marks]

Worked method: Expected output [1]; The result that should occur according to the problem requirements. [1]. Refinement [1]; Use discrepancies in the trace to revise instructions and retest. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the total printed by this loop.

```
total = 0
for i = 1 to 3
    total = total + i
next i
print(total)
```

predict

Question 1

State TWO concepts relevant to this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 2

Explain how accumulator and expected output help in a sports count. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 3

Explain the roles of trace table, expected output and refinement in a fundraising sum. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A trace table records changes to variable values and outputs as instructions execute.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The input, expected output, actual output or identified fault, with the correction.

Q2. D — Only those they need

Q3. It enables communication with and control of a particular peripheral.

Q4. (a) readLine() reads the next line and moves forward. (b) writeLine(text) writes a line to an open file. Open, read or write, then close. Screen output does not save a file.

I Do / We Do pages · Guided practice

Q1. Trace table [1]; Manually records changes to variables and printed outputs as code executes. [1].

Q2. Accumulator [1]; A variable updated repeatedly to store a running total. [1].

Q3. Expected output [1]; The result that should occur according to the problem requirements. [1]. Refinement [1]; Use discrepancies in the trace to revise instructions and retest. [1].

You Do pages · Code First

```
total = 0
for i = 1 to 3
    total = total + i
next i
print(total)
```

Expected result: 6

You Do pages · Independent questions

Q1. Trace table [1]; Expected output [1].

Q2. Accumulator [1]; A variable updated repeatedly to store a running total. [1]. Expected output [1]; The result that should occur according to the problem requirements. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Trace table [1]; Manually records changes to variables and printed outputs as code executes. [1]. Expected output [1]; The result that should occur according to the problem requirements. [1]. Refinement [1]; Use discrepancies in the trace to revise instructions and retest. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: